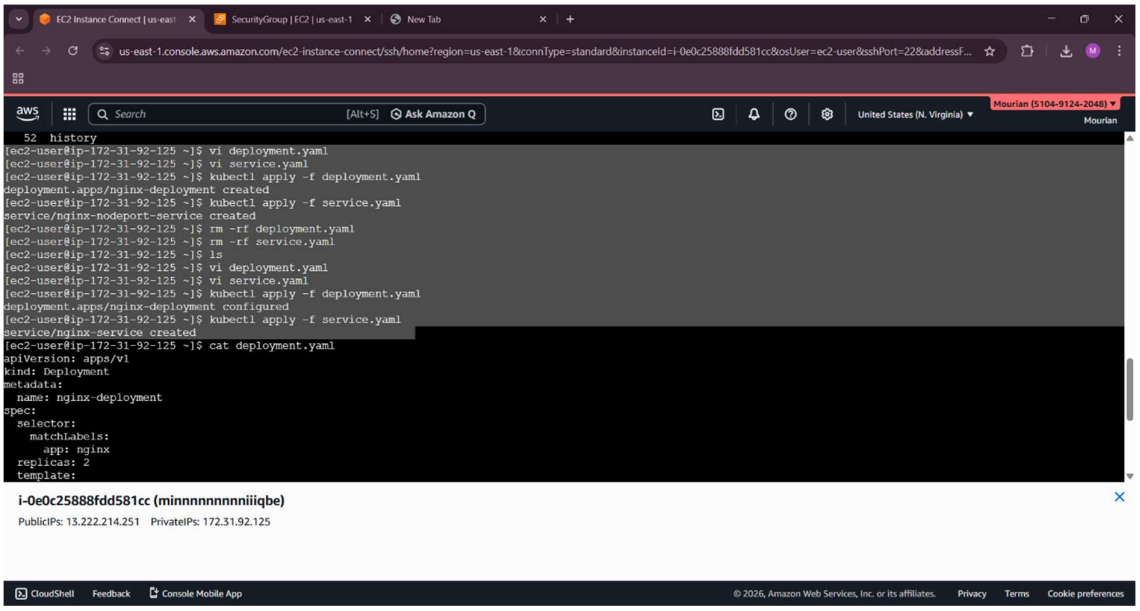
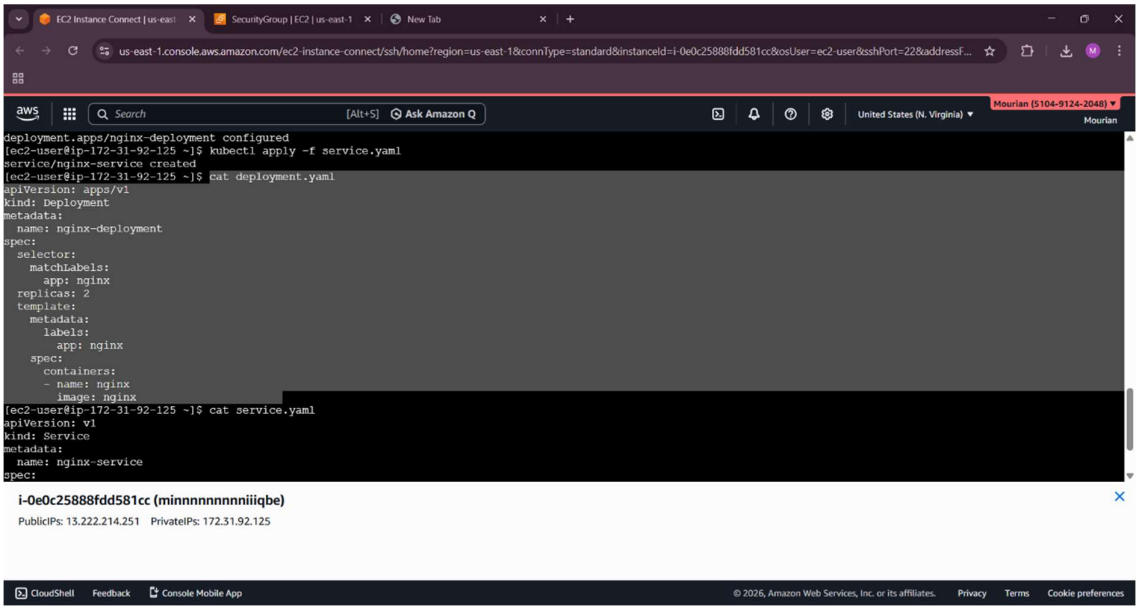
ASSIGNMENT 41

Kubernetes- load balancer





```
  name: nginx-deployment
spec:
  selector:
    matchLabels:
      app: nginx
  replicas: 2
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx
[ec2-user@ip-172-31-92-125 ~]$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: LoadBalancer
  selector:
    app: nginx
  ports:
    - port: 80
      targetPort: 80
[ec2-user@ip-172-31-92-125 ~]$
```

i-0e0c25888fdd581cc (minnnnnnnnniiiqbe)

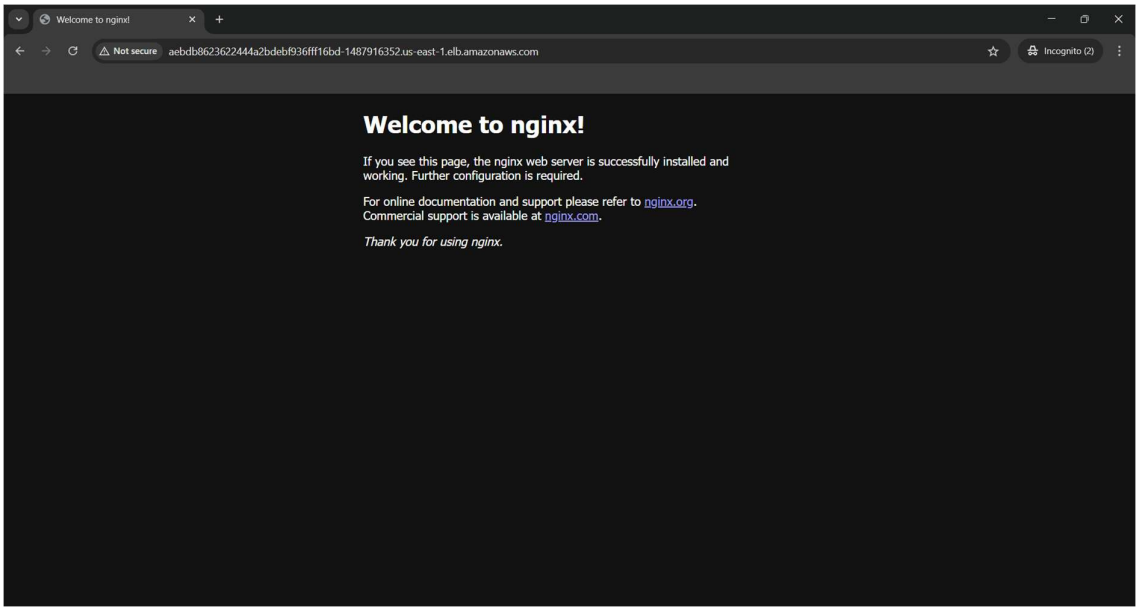
PublicIPs: 13.222.214.251 PrivateIPs: 172.31.92.125

```
-bash: minikubu: command not found
[ec2-user@ip-172-31-92-125 ~]$ minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
[ec2-user@ip-172-31-92-125 ~]$ aws configure
AWS Access Key ID [None]: AKIAXNW5GFZAOQ7VN7V5
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
[ec2-user@ip-172-31-92-125 ~]$ aws eks --region us-east-1 update-kubeconfig --name eksneww1222
Added new context arn:aws:eks:us-east-1:510491242048:cluster/eksneww1222 to /home/ec2-user/.kube/config
[ec2-user@ip-172-31-92-125 ~]$ kubectl get nodes
NAME                                       STATUS   ROLES    AGE   VERSION
ip-172-31-15-67.ec2.internal               Ready    <none>   15m   v1.35.0-eks-efcacff
ip-172-31-94-254.ec2.internal              Ready    <none>   15m   v1.35.0-eks-efcacff
[ec2-user@ip-172-31-92-125 ~]$ history
    1  sudo dnf update -y
    2  sudo dnf install -y curl wget tar conntrack
    3  sudo dnf install -y docker
    4  sudo systemctl enable docker
    5  sudo systemctl start docker
    6  sudo usermod -aG docker ec2-user
    7  newgrp docker
    8  curl -LO https://github.com/kubernetes/minikube/releases/latest/download/minikube-linux-amd64
    9  sudo install minikube-linux-amd64 /usr/local/bin/minikube && rm minikube-linux-amd64
   10  minikube --version
   11  minikube version
```

i-0e0c25888fdd581cc (minnnnnnnnniiiqbe)

PublicIPs: 13.222.214.251 PrivateIPs: 172.31.92.125

This output from EKS two instance nginx web hosting by using load balancer ：



Deployment.yaml:

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  selector:
    matchLabels:
      app: nginx
  replicas: 2
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx
```

Service.yaml:

```yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: LoadBalancer
  selector:
    app: nginx
  ports:
    - port: 80
      targetPort: 80
```